

Example Use the algebra of limits theorem to find $\lim_{n \rightarrow \infty} x_n$

where

$$x_n = \frac{2}{3} \frac{2n^2 - n}{-3n^2 + 2}$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(\frac{2}{3} \frac{2n^2 - n}{-3n^2 + 2} \right)$$

$$= \frac{2}{3} \lim_{n \rightarrow \infty} \left(\frac{2n^2 - n}{-3n^2 + 2} \right)$$

(can't apply the quotient case yet, as numerator and denominator are divergent)

$$= \frac{2}{3} \lim_{n \rightarrow \infty} \left(\frac{2 - \frac{1}{n}}{-3 + \frac{2}{n^2}} \right), \text{ div. by } \frac{1}{n^2}$$

$$= \frac{2}{3} \frac{\lim_{n \rightarrow \infty} (2 - \frac{1}{n})}{\lim_{n \rightarrow \infty} (-3 + \frac{2}{n^2})}, \text{ quotient case of algebra of limits theorem.}$$

$$= \frac{2}{3} \frac{2 - \lim(\frac{1}{n})}{-3 + 2 \lim(\frac{1}{n^2})}, \text{ by lin. combs of alg. of lim. th.}$$

(already proved that $\frac{1}{n}, \frac{1}{n^2} \rightarrow 0$
as $n \rightarrow \infty$).

$$= \frac{2}{3} \cdot \frac{2}{-3}$$

$$= -4/9.$$

$|a_n - a|$
gets
small

$|b_n - b|$
gets
small

Proof of lin. combs. case

let $a_n \rightarrow a$, $b_n \rightarrow b$ as $n \rightarrow \infty$
and let $\alpha, \beta \neq 0$.

Claim: $\lim_{n \rightarrow \infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b$

Investigate the final inequality we
have to establish.

$$|(\alpha a_n + \beta b_n) - (\alpha a + \beta b)| < \varepsilon.$$

$$\Leftrightarrow |\alpha(a_n - a) + \beta(b_n - b)| < \varepsilon$$

Note that

$$\begin{aligned} & |\alpha(a_n - a) + \beta(b_n - b)| \\ & \leq |\alpha(a_n - a)| + |\beta(b_n - b)| \end{aligned}$$

Want

$$\frac{a_n}{b_n} \rightarrow \frac{a}{b} = a \left(\frac{1}{b} \right).$$

Th. 14.7 Monotone Convergence Theorem

Every bounded monotonic sequence is convergent

eg. if a sequence $\{a_n\}$ is bounded above and increasing then it is convergent; so $\exists a \in \mathbb{R}$

such that $a_n \rightarrow a$ $\{b_n\}$

OR if a sequence $\{b_n\}$ is bounded below and decreasing then it is convergent, i.e. $\exists b \in \mathbb{R}$ and $b_n \rightarrow b$ as $n \rightarrow \infty$.

Q8 From tutorial sheet.

$$\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots$$

a) $a_1 = \sqrt{2}$

and for all $n \geq 1$

$$a_{n+1} = \sqrt{2a_n}$$

Firstly let's prove $\{a_n\}$ is bounded above by 2.

Use induction.

Base case $a_1 = \sqrt{2} < 2$. ✓

Now assume that $a_k < 2$, for $k \geq 1$.

$$\begin{aligned} a_{k+1} &= \sqrt{2a_k} \\ &< \sqrt{2 \cdot 2} \\ &= 2. \end{aligned}$$

So by induction, $\forall n \geq 1, a_n < 2$.

Now let's prove $\{a_n\}$ is increasing.

$$\Rightarrow a = \sqrt{2} \lim_{n \rightarrow \infty} (\sqrt{a_n})$$

$$\Rightarrow a = \sqrt{2} \sqrt{a}, \quad \text{since sqrt. is a continuous function.}$$

$$\Rightarrow a^2 = 2a.$$

$$\Rightarrow \boxed{a = 2.}$$

Q9]

~~$$a_0 = 1$$~~

$$a_0 = 1 \quad \forall n \geq 1 \quad a_n = 3 - \frac{1}{a_{n-1}}$$

Prove $\{a_n\}$ is increasing and bounded.

First let's prove sequence is bounded below by 1.

Base case $a_0 = 1 \geq 1$

Assume $a_n \geq 1$ then.

$$0 < \frac{1}{a_n} \leq 1.$$

